# EXPERIMENT – 5

**AIM :**

To create SQL views for department salary summary and employee hierarchy, test the updatability of views, and implement a recursive CTE to display reporting chains.

**SOURCE CODE :**

```
CREATE TABLE Department (
   Dept_ID INT PRIMARY KEY,
   Dept_Name VARCHAR(50),
   Location VARCHAR(50)
);

CREATE TABLE Project (
   Project_ID INT PRIMARY KEY,
   Project_Name VARCHAR(100),
   Budget DECIMAL(12,2)
);

CREATE TABLE Employee (
   Emp_ID INT PRIMARY KEY,
   Emp_Name VARCHAR(50),
   Salary DECIMAL(10,2),
   Job_Role VARCHAR(50),
   Dept_ID INT,
   Project_ID INT,
   Manager_ID INT,
   FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID),
   FOREIGN KEY (Project_ID) REFERENCES Project(Project_ID)
);

INSERT INTO Department VALUES
(1, 'Human Resources', 'Delhi'),
(2, 'Information Technology', 'Chandigarh'),
(3, 'Finance', 'Mumbai'),
(4, 'Marketing', 'Bangalore'),
(5, 'Research and Development', 'Hyderabad');

INSERT INTO Project VALUES
(101, 'Website Development', 500000),
(102, 'Mobile Application', 750000),
(103, 'Cyber Security System', 900000),
(104, 'Data Analytics Platform', 850000),
(105, 'AI Chatbot', 1200000),
(106, 'Cloud Migration', 1000000),
(107, 'Payroll Management', 400000),
(108, 'Market Research System', 600000);
```

```
INSERT INTO Employee VALUES
(1, 'Aarav Sharma', 45000, 'HR Executive', 1, 107, NULL),
(2, 'Ananya Gupta', 52000, 'HR Manager', 1, 107, NULL),
(3, 'Riya Verma', 48000, 'Recruiter', 1, 107, 2),
(4, 'Karan Singh', 55000, 'HR Analyst', 1, 107, 2),
(5, 'Priya Mehta', 60000, 'HR Manager', 1, 107, NULL),
(6, 'Rahul Kumar', 75000, 'Software Engineer', 2, 101, 10),
(7, 'Neha Sharma', 82000, 'Software Engineer', 2, 102, 10),
(8, 'Aditya Singh', 95000, 'Security Engineer', 2, 103, 10),
(9, 'Simran Kaur', 88000, 'Data Analyst', 2, 104, 10),
(10, 'Vivek Gupta', 105000, 'Cloud Engineer', 2, 106, NULL),
(11, 'Ishita Jain', 78000, 'Software Developer', 2, 101, 10),
(12, 'Rohan Verma', 92000, 'AI Engineer', 2, 105, 10),
(13, 'Aman Yadav', 65000, 'Financial Analyst', 3, 107, 15),
(14, 'Sneha Kapoor', 72000, 'Accountant', 3, 107, 15),
(15, 'Mohit Agarwal', 85000, 'Finance Manager', 3, 107, NULL),
(16, 'Pooja Sharma', 68000, 'Financial Analyst', 3, 107, 15),
(17, 'Nikhil Jain', 90000, 'Senior Accountant', 3, 107, 15),
(18, 'Kriti Verma', 62000, 'Accountant', 3, 107, 15),
(19, 'Arjun Malhotra', 58000, 'Marketing Executive', 4, 108, 20),
(20, 'Tanya Singh', 70000, 'Marketing Manager', 4, 108, NULL),
(21, 'Yash Gupta', 62000, 'Marketing Analyst', 4, 108, 20),
(22, 'Muskan Sharma', 75000, 'Brand Manager', 4, 108, 20),
(23, 'Dev Kumar', 66000, 'Sales Executive', 4, 108, 20),
(24, 'Naina Kapoor', 80000, 'Marketing Analyst', 4, 108, 20),
(25, 'Kabir Sharma', 95000, 'Research Scientist', 5, 105, 28),
(26, 'Meera Gupta', 88000, 'ML Engineer', 5, 105, 28),
(27, 'Varun Singh', 92000, 'Research Analyst', 5, 104, 28),
(28, 'Aditi Verma', 110000, 'AI Engineer', 5, 105, NULL),
(29, 'Harsh Jain', 98000, 'Data Scientist', 5, 104, 28),
(30, 'Isha Kapoor', 85000, 'Research Engineer', 5, 103, 28);

CREATE VIEW Department_Salary_Summary AS
SELECT
   d.Dept_ID,
   d.Dept_Name,
   COUNT(e.Emp_ID) AS Employee_Count,
   AVG(e.Salary) AS Average_Salary,
   SUM(e.Salary) AS Total_Salary,
   MAX(e.Salary) AS Highest_Salary,
   MIN(e.Salary) AS Lowest_Salary
FROM Department d
LEFT JOIN Employee e
ON d.Dept_ID = e.Dept_ID
GROUP BY d.Dept_ID, d.Dept_Name;

SELECT * FROM Department_Salary_Summary;

CREATE VIEW Employee_Hierarchy AS
SELECT
```

```
    e.Emp_ID,
    e.Emp_Name,
    e.Job_Role,
    e.Dept_ID,
    e.Manager_ID,
    m.Emp_Name AS Manager_Name
FROM Employee e
LEFT JOIN Employee m
ON e.Manager_ID = m.Emp_ID;

SELECT * FROM Employee_Hierarchy;

CREATE VIEW Employee_Basic_View AS
SELECT
    Emp_ID,
    Emp_Name,
    Salary,
    Dept_ID
FROM Employee;

UPDATE Employee_Basic_View
SET Salary = Salary + 1000
WHERE Emp_ID = 1;

SELECT * FROM Employee_Basic_View
WHERE Emp_ID = 1;

WITH RECURSIVE Reporting_Chain AS
(
    SELECT
        Emp_ID,
        Emp_Name,
        Manager_ID,
        0 AS Level
    FROM Employee
    WHERE Manager_ID IS NULL

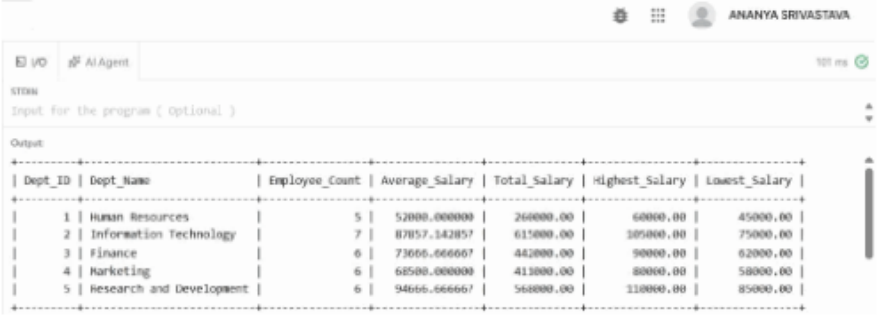
    UNION ALL

    SELECT
        e.Emp_ID,
        e.Emp_Name,
        e.Manager_ID,
        rc.Level + 1
    FROM Employee e
    INNER JOIN Reporting_Chain rc
    ON e.Manager_ID = rc.Emp_ID
)
SELECT
    Emp_ID,
```

```
    Emp_Name,
    Manager_ID,
    Level
FROM Reporting_Chain
ORDER BY Level, Emp_ID;
```

## OUTPUT :

### Department_Salary_Summary



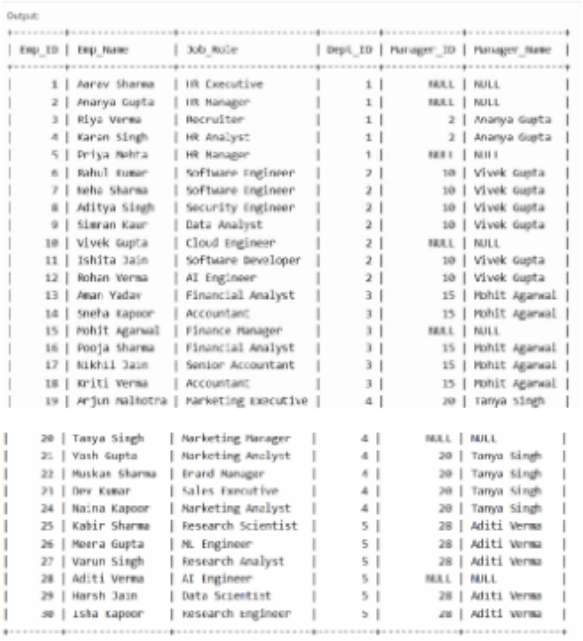
```
ANANYA SRIVASTAVA

I/O   AI Agent                                                    101 ms

STDIN
Input for the program ( Optional )

Output:
+---------+-------------------------+----------------+-----------------+--------------+----------------+---------------+
| Dept_ID | Dept_Name               | Employee_Count | Average_Salary  | Total_Salary | Highest_Salary | Lowest_Salary |
+---------+-------------------------+----------------+-----------------+--------------+----------------+---------------+
|       1 | Human Resources         |              5 |   52000.000000  |    260000.00 |       60000.00 |      45000.00 |
|       2 | Information Technology  |              7 |   87857.142857  |    615000.00 |      105000.00 |      75000.00 |
|       3 | Finance                 |              6 |   73666.666667  |    442000.00 |       90000.00 |      62000.00 |
|       4 | Marketing               |              6 |   68500.000000  |    411000.00 |       80000.00 |      58000.00 |
|       5 | Research and Development |             6 |   94666.666667  |    568000.00 |      110000.00 |      85000.00 |
+---------+-------------------------+----------------+-----------------+--------------+----------------+---------------+
```

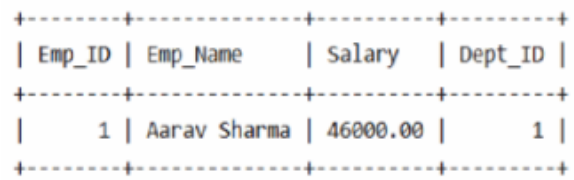
### Employee_Hierarchy



```
Output:
+--------+---------------+---------------------+---------+-------------+-----------------+
| Emp_ID | Emp_Name      | Job_Role            | Dept_ID | Manager_ID  | Manager_Name    |
+--------+---------------+---------------------+---------+-------------+-----------------+
|      1 | Aarav Sharma  | HR Executive        |       1 |        NULL | NULL            |
|      2 | Ananya Gupta  | HR Manager          |       1 |        NULL | NULL            |
|      3 | Riya Verma    | Recruiter           |       1 |           2 | Ananya Gupta    |
|      4 | Karan Singh   | HR Analyst          |       1 |           2 | Ananya Gupta    |
|      5 | Priya Mehta   | HR Manager          |       1 |        NULL | NULL            |
|      6 | Rahul Kumar   | Software Engineer   |       2 |          10 | Vivek Gupta     |
|      7 | Neha Sharma   | Software Engineer   |       2 |          10 | Vivek Gupta     |
|      8 | Aditya Singh  | Security Engineer   |       2 |          10 | Vivek Gupta     |
|      9 | Simran Kaur   | Data Analyst        |       2 |          10 | Vivek Gupta     |
|     10 | Vivek Gupta   | Cloud Engineer      |       2 |        NULL | NULL            |
|     11 | Ishita Jain   | Software Developer  |       2 |          10 | Vivek Gupta     |
|     12 | Rohan Verma   | AI Engineer         |       2 |          10 | Vivek Gupta     |
|     13 | Aman Yadav    | Financial Analyst   |       3 |          15 | Mohit Agarwal   |
|     14 | Sneha Kapoor  | Accountant          |       3 |          15 | Mohit Agarwal   |
|     15 | Mohit Agarwal | Finance Manager     |       3 |        NULL | NULL            |
|     16 | Pooja Sharma  | Financial Analyst   |       3 |          15 | Mohit Agarwal   |
|     17 | Nikhil Jain   | Senior Accountant   |       3 |          15 | Mohit Agarwal   |
|     18 | Kriti Verma   | Accountant          |       3 |          15 | Mohit Agarwal   |
|     19 | Arjun Malhotra | Marketing Executive |      4 |          20 | Tanya Singh     |

|     20 | Tanya Singh   | Marketing Manager   |       4 |        NULL | NULL            |
|     21 | Yash Gupta    | Marketing Analyst   |       4 |          20 | Tanya Singh     |
|     22 | Muskan Sharma | Brand Manager       |       4 |          20 | Tanya Singh     |
|     23 | Dev Kumar     | Sales Executive     |       4 |          20 | Tanya Singh     |
|     24 | Naina Kapoor  | Marketing Analyst   |       4 |          20 | Tanya Singh     |
|     25 | Kabir Sharma  | Research Scientist  |       5 |          28 | Aditi Verma     |
|     26 | Meera Gupta   | ML Engineer         |       5 |          28 | Aditi Verma     |
|     27 | Varun Singh   | Research Analyst    |       5 |          28 | Aditi Verma     |
|     28 | Aditi Verma   | AI Engineer         |       5 |        NULL | NULL            |
|     29 | Harsh Jain    | Data Scientist      |       5 |          28 | Aditi Verma     |
|     30 | Isha Kapoor   | Research Engineer   |       5 |          28 | Aditi Verma     |
+--------+---------------+---------------------+---------+-------------+-----------------+
```

### Employee_Basic_View



```
+--------+--------------+----------+---------+
| Emp_ID | Emp_Name     | Salary   | Dept_ID |
+--------+--------------+----------+---------+
|      1 | Aarav Sharma | 46000.00 |       1 |
+--------+--------------+----------+---------+
```

**Reporting_Chain**

Output:

| Emp_ID | Emp_Name | Manager_ID | Level |
|---|---|---|---|
| 1 | Aarav Sharma | NULL | 0 |
| 2 | Ananya Gupta | NULL | 0 |
| 5 | Priya Mehta | NULL | 0 |
| 10 | Vivek Gupta | NULL | 0 |
| 15 | Mohit Agarwal | NULL | 0 |
| 20 | Tanya Singh | NULL | 0 |
| 28 | Aditi Verma | NULL | 0 |
| 3 | Riya Verma | 2 | 1 |
| 4 | Karan Singh | 2 | 1 |
| 6 | Rahul Kumar | 10 | 1 |
| 7 | Neha Sharma | 10 | 1 |
| 8 | Aditya Singh | 10 | 1 |
| 9 | Simran Kaur | 10 | 1 |
| 11 | Ishita Jain | 10 | 1 |
| 12 | Rohan Verma | 10 | 1 |
| 13 | Aman Yadav | 15 | 1 |
| 14 | Sneha Kapoor | 15 | 1 |
| 16 | Pooja Sharma | 15 | 1 |
| 17 | Nikhil Jain | 15 | 1 |
| 18 | Kriti Verma | 15 | 1 |
| 19 | Arjun Malhotra | 20 | 1 |
| 21 | Yash Gupta | 20 | 1 |
| 22 | Muskan Sharma | 20 | 1 |
| 23 | Dev Kumar | 20 | 1 |
| 24 | Naina Kapoor | 20 | 1 |
| 25 | Kabir Sharma | 28 | 1 |
| 26 | Meera Gupta | 28 | 1 |
| 27 | Varun Singh | 28 | 1 |
| 29 | Harsh Jain | 28 | 1 |
| 30 | Isha Kapoor | 28 | 1 |